

Ensemble Techniques

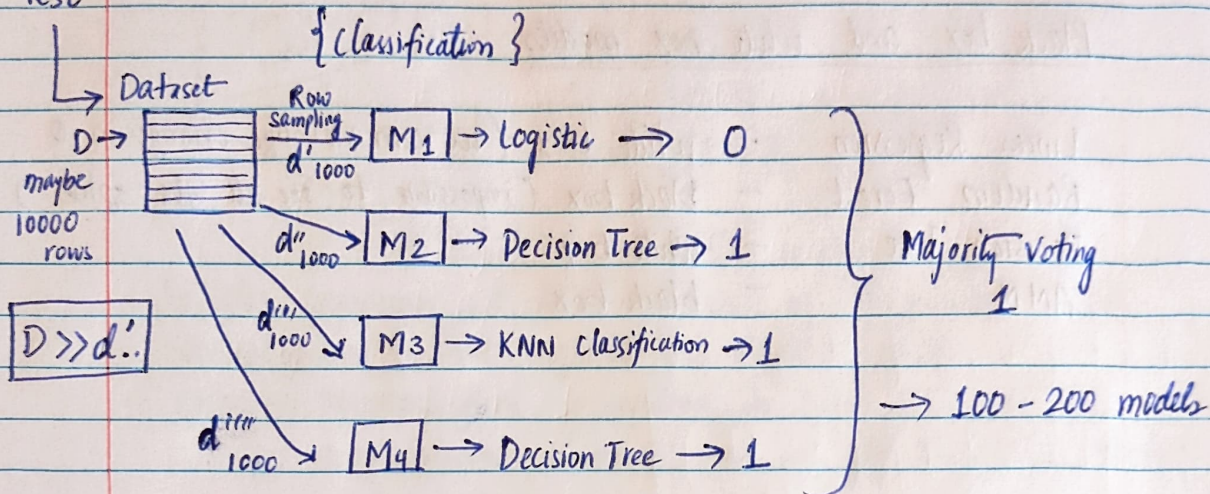
Multiple algorithms to solve a problem?

Ensemble Techniques

Bagging

Boosting

Test



This step is called as Bootstrap Aggregator, there won't be any tie between the models having 50% yes and 50% no. We use many models together.

W.r.t regression problem let say for $M_1 \rightarrow 120$

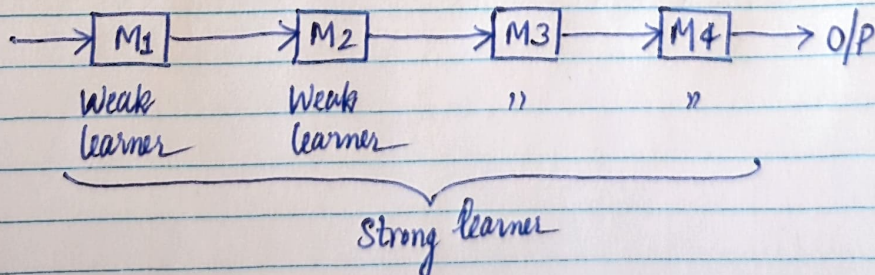
$M_2 \rightarrow 140$

$M_3 \rightarrow 122$

$M_4 \rightarrow 148$

mean is taken as o/p of the regression

Boosting



Bagging

- 1> Random forest classifier
- 2> Random forest Regressor

Boosting

- 1> Adaboost
- 2> Gradient boost
- 3> XGboost

Black box and white box models

- | | |
|-------------------|--|
| Linear Regression | - white box (we can visualize change in θ) |
| Random Forest | - black box (impossible to see all the splits) |
| Decision Tree | - white box |
| ANN | - black box |